Experiment No: 05

Name of The Experiment: Verification of Kirchhoff's Voltage Law and Kirchhoff's Current Law.

Objectives:

a) To verify Kirchhoff's Voltage Law.
b) To verify Kirchhoff's Current Law.

Theory:

Kirchhoff's Voltage Law

Kirchhoff's Voltage Law (KVL) states that the algebraic sum of all branch voltages around any closed path in a circuit is always zero at all instants of time. In the figure 5(a), if KVL is applied then the equation is

$$V = V_1+V_2+V_3$$

Kirchhoff's Current Law

Kirchhoff's Current Law (KCL) states that the sum of the currents entering into any node/point/junction is equal to the sum of the currents leaving that node/point/junction. In the figure 5(b), if KCL is applied then the equation is

$$I = I_1 + I_2$$

Theoretical Diagram:

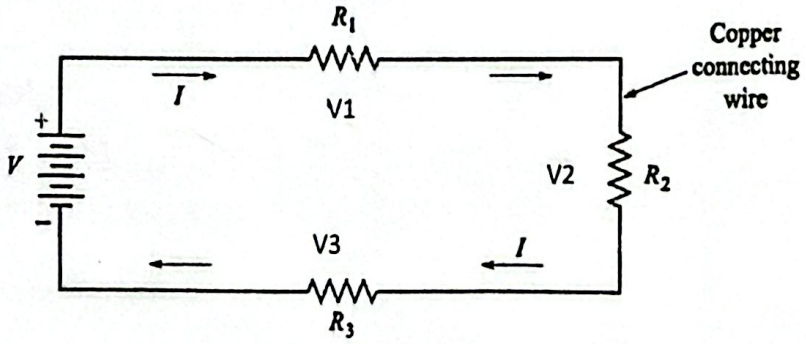


Fig 5(a): Circuit for KVL

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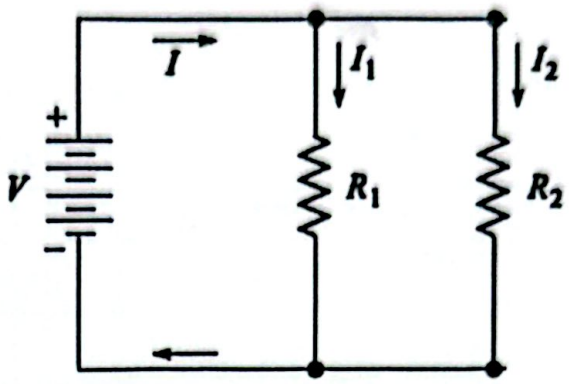


Fig 5(b): Circuit for KCL

Apparatus:

a) DC power supply
b) Breadboard
c) Resistors of various values
d) Voltmeter
e) Ammeter
f) Connecting wires

Procedure:

Verification of KVL

a) Connection are made as per the circuit diagram shown in fig 5(a).
b) Vary the supply voltage and take the corresponding readings V1 & V2 from the voltmeter.
c) Verify the reading.

Verification of KCL

a) Give the connection according to circuit shown in fig 5(b).
b) Vary the supply voltage and take the corresponding readings of I, I_1 & I_2 from the ammeter.
c) Verify the reading.

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Experimental Table:

Data Table 01: Verification of KVL

OBS	V(V)	V_1(V)		V_2(V)		V_3(V)		$V = V_1+V_2+V_3$ (V)	
		Cal	Mea	Cal	Mea	Cal	Mea	Cal	Mea
1.									
2.									
3.									

Data Table 02: Verification of KCL

OBS	V(V)	I_1(A)		I_2(A)		$I = I_1 + I_2$	
		Cal	Mea	Cal	Mea	Cal	Mea
1.							
2.							
3.							

Calculations:

KVL

OBS 1:

$$V1 = \frac{V \times R1}{R1+R2+R3} \text{ (Volt) .....[calculated value]}$$

$$V2 = \frac{V \times R2}{R1+R2+R3} \text{ (Volt) .....[calculated value]}$$

$$V3 = \frac{V \times R3}{R1+R2+R3} \text{ (Volt) .....[calculated value]}$$

$V = V_1+V_2+V_3$ .....[measured value]

$V = V_1+V_2+V_3$ .....[calculated value]

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OBS 2:

$$V1 = \frac{V \times R1}{R1+R2+R3} \text{ (Volt) .....[calculated value]}$$

$$V2 = \frac{V \times R2}{R1+R2+R3} \text{ (Volt) .....[calculated value]}$$

$$V3 = \frac{V \times R3}{R1+R2+R3} \text{ (Volt) .....[calculated value]}$$

$V = V_1+V_2+V_3$ .....[measured value]

$V = V_1+V_2+V_3$ .....[calculated value]

OBS 3:

$$V1 = \frac{V \times R1}{R1+R2+R3} \text{ (Volt) .....[calculated value]}$$

$$V2 = \frac{V \times R2}{R1+R2+R3} \text{ (Volt) .....[calculated value]}$$

$$V3 = \frac{V \times R3}{R1+R2+R3} \text{ (Volt) .....[calculated value]}$$

$V = V_1+V_2+V_3$ .....[measured value]

$V = V_1+V_2+V_3$ .....[calculated value]

KCL

OBS 1:

$$I_1 = \frac{I \times R2}{R1+R2} \text{ (A) .....[calculated value]}$$

$$I_2 = \frac{I \times R1}{R1+R2} \text{ (A) .....[calculated value]}$$

$I = I_1 + I_2$ .....[measured value]

$I = I_1 + I_2$.....[calculated value]

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OBS 2:

$I_1 = \frac{I \times R2}{R1+R2}$ (A) .....[calculated value]

$I_2 = \frac{I \times R1}{R1+R2}$ (A) .....[calculated value]

$I = I_1 + I_2$ .....[measured value]

$I = I_1 + I_2$.....[calculated value]

OBS 3:

$I_1 = \frac{I \times R2}{R1+R2}$ (A) .....[calculated value]

$I_2 = \frac{I \times R1}{R1+R2}$ (A) .....[calculated value]

$I = I_1 + I_2$ .....[measured value]

$I = I_1 + I_2$.....[calculated value]

Results:

KVL:

a) OBS 1: Measured Value of Voltage.....-- Calculated Value of Voltage .....

b) OBS 2: Measured Value of Voltage .....-- Calculated Value of Voltage .....

c) OBS 3: Measured Value of Voltage .....-- Calculated Value of Voltage .....

KCL:

a) OBS 1: Measured Value of Current.....-- Calculated Value of Current .....

b) OBS 2: Measured Value of Current .....-- Calculated Value of Current .....

c) OBS 3: Measured Value of Current .....-- Calculated Value of Current .....

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Discussion:

The measured values and calculated values are nearly same. The difference between the is for instrumental and observational error. Neglecting this error Kirchhoff's Voltage Law and Kirchhoff's Current Law are verified successfully.

N.B: Fluctuations in temperature, humidity, and other environmental conditions can affect the performance of components and measurement instruments.

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